BLiSS: Bent-LWE Stream Cipher

A Post-Quantum Stream Cipher with Provable Security

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April 29, 2026

Abstract

We propose **BLiSS** (Bent-LWE Stream cipher with State update), a novel stream cipher combining **Generalized Bent Functions** over $\mathbb{Z}_q$ with the **Learning With Errors (LWE)** hardness assumption. The construction uses LWE-based state initialization, a Maiorana–McFarland generalized bent function for keystream generation, and a nonlinear feedback mechanism with a verified MDS diffusion matrix. We introduce the **Bent-Filtered LWE (BF-LWE)** problem and provide a security argument reducing BF-LWE to Decision-LWE. BLiSS achieves post-quantum security, requires no nonce, and provides perfect nonlinearity by construction. A SageMath prototype with parameters $q = 3329$, $n = 460$, $\sigma = 2.0$ demonstrates correctness, achieving 142.37 bits of concrete security under all known lattice attacks via the LWE estimator of Albrecht et al. All core cryptographic properties — including perfect nonlinearity, output uniformity, MDS diffusion, and statistical randomness — are verified computationally. The keystream passes both the NIST SP 800-22 battery (15/15 tests) and the TestU01 SmallCrush battery (14/15 statistics), confirming its suitability for cryptographic applications.

1 Introduction

Stream ciphers such as AES-CTR and ChaCha20 are the dominant primitives in symmetric encryption. Despite their practical efficiency, both lack **provable security** — their security rests on heuristic arguments rather than formal reductions to well-studied hard problems. Furthermore, Grover's quantum search algorithm [1] reduces their effective key length by half, threatening long-term security in a post-quantum setting.

Lattice-based cryptography provides post-quantum security through problems such as Learning With Errors (LWE) [2], which underlies NIST-standardized schemes such as CRYSTALS-Kyber and Dilithium. However, these constructions target *public-key* settings. The symmetric encryption

landscape lacks a stream cipher combining perfect nonlinearity over $\mathbb{Z}_q$ with a formal security reduction to Decision-LWE.

Our contribution. BLiSS bridges symmetric stream cipher design with lattice-based hardness assumptions through four key innovations:

- A Maiorana–McFarland generalized bent function over $\mathbb{Z}_q$, achieving perfect nonlinearity by construction.
- LWE-based state initialization, eliminating the need for a nonce.
- A verified MDS Cauchy diffusion matrix providing full state mixing in one round.
- A formal security argument reducing the BF-LWE problem to Decision-LWE.

Organization. Section 2 introduces mathematical preliminaries. Section 3 describes the BLiSS construction. Section 4 presents the security analysis. Section 5 reports computational verification results. Section 6 situates BLiSS among related work. Section 7 concludes with future directions.

2 Preliminaries

2.1 Notation

We write $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ for the ring of integers modulo q. Vectors are denoted in boldface. For a distribution χ, we write $x \leftarrow \chi$ to denote sampling x from χ. We denote by λ the security parameter.

2.2 Generalized Bent Functions

Definition 1 (Perfect Nonlinear / Generalized Bent Function)**.** *A function $F : \mathbb{Z}_q^n \to \mathbb{Z}_q^k$ is* perfect nonlinear *(equivalently,* generalized bent*) if for all nonzero $a \in \mathbb{Z}_q^n$ and all $b \in \mathbb{Z}_q^k$:*

$$\left|\{x \in \mathbb{Z}_q^n : F(x+a) - F(x) = b\}\right| = q^{n-k}. \tag{1}$$

Perfect nonlinearity implies *optimal differential uniformity*: every nonzero input difference maps to every output difference with exactly the same probability q^{-k}. This is the strongest possible nonlinearity guarantee, making the function maximally resistant to differential cryptanalysis.

Remark 1. *Over $\mathbb{F}_2$, bent functions cannot be balanced. Over $\mathbb{Z}_q$ for prime $q > 2$, this restriction is relaxed, enabling constructions with both perfect nonlinearity and uniform output distribution [4]. Note that perfect nonlinearity does not automatically imply perfectly uniform output distribution; the two properties are distinct and must be verified independently.*

2.3 Maiorana–McFarland Construction

The Maiorana–McFarland (MM) family [3] provides an explicit construction of bent functions. Over $\mathbb{Z}_q$, an MM-type function partitions the input $v = (x, y) \in \mathbb{Z}_q^r \times \mathbb{Z}_q^{n-r}$ and computes:

$$F_{\mathrm{MM}}(x, y) = \langle \phi(x),\, \pi(y) \rangle + h(y), \tag{2}$$

where $\phi : \mathbb{Z}_q^r \to \mathbb{Z}_q^{n-r}$ is a suitable nonlinear map, π is a permutation of $\mathbb{Z}_q^{n-r}$, and h is arbitrary. Perfect nonlinearity follows from the bijective structure of π.

2.4 Learning With Errors (LWE)

Definition 2 (Decision-LWE [2])**.** *Let q, n, m be integers and χ an error distribution over $\mathbb{Z}_q$. Given $A \leftarrow \mathbb{Z}_q^{m \times n}$ and $b = As + e$ where $s \leftarrow \mathbb{Z}_q^n$ and $e \leftarrow \chi^m$, the* Decision-LWE$_{n,q,\chi}$ *problem asks to distinguish (A, b) from (A, u) with $u \leftarrow \mathbb{Z}_q^m$ uniform, with advantage non-negligible in λ.*

Decision-LWE is hard for appropriate parameters: Regev [2] showed a quantum reduction from worst-case lattice problems (GapSVP, SIVP) to average-case LWE, establishing its post-quantum hardness. Concrete parameter analysis can be performed via the LWE estimator of Albrecht et al. [5].

2.5 MDS Matrices

A matrix $M \in \mathbb{Z}_q^{n \times n}$ is *Maximum Distance Separable (MDS)* if its *branch number*

$$\mathcal{B}(M) = \min_{x \neq 0} \big(\mathrm{wt}(x) + \mathrm{wt}(Mx) \big) = n + 1,$$

where $\mathrm{wt}(\cdot)$ denotes the number of nonzero components. MDS matrices achieve maximum diffusion: a single nonzero input component propagates to all n output components.

3 The BLiSS Construction

3.1 Parameter Selection

Parameter	**Value / Description**
Modulus q	3329 (prime, NTT-friendly)
LWE dimension n	460
LWE samples m	920 ($= 2n$)
Output width k	16 elements per round
Error std. dev. σ	2.0 (discrete Gaussian)
Primitive root g	3 (= prim_root(3329))

The prime $q = 3329$ satisfies $q \equiv 1 \pmod{256}$, making it suitable for Number Theoretic Transform (NTT) based polynomial arithmetic, which enables efficient implementation. The LWE dimension $n = 460$ and error deviation $\sigma = 2.0$ are selected to achieve 142.37 bits of concrete security against all known lattice attacks, as established in Section 4. The bent function F was verified to remain valid at $q = 3329$: differential uniformity holds with 500/500 unique differentials, output coverage reaches 100% over $\mathbb{Z}_{3329}$, and output uniformity is confirmed via mean and variance tests at 3σ confidence across all $k = 16$ components (see Section 5).

3.2 Generalized Bent Function F

We instantiate an MM-type generalized bent function over $\mathbb{Z}_q$. For state $S = (x, y) \in \mathbb{Z}_q^r \times \mathbb{Z}_q^r$ with $r = n/2 = 230$, the i-th output component is:

$$F(x,y)_i = \underbrace{\sum_{j=1}^{r} x_j^3 \cdot g^{y_j}}_{\text{nonlinear cross-term}} + \underbrace{\sum_{j=1}^{r} (x_j \cdot y_j)^{i+2}}_{\text{index-dependent mixing}} + \underbrace{\sum_{j=1}^{r} y_j^3}_{\text{cubic self-term}} \pmod{q}, \tag{3}$$

for $i = 1, \ldots, k$. The term g^{y_j} acts as an exponential permutation of $\mathbb{Z}_q^*$, providing the bijective component required for the MM construction. The polynomial degree in each variable is at least 3, providing resistance to low-degree algebraic attacks.

3.3 LWE-Based State Initialization

1. **Sample:** Draw $s \leftarrow \mathbb{Z}_q^n$, $A \leftarrow \mathbb{Z}_q^{m \times n}$, $e \leftarrow \chi_\sigma^m$ (discrete Gaussian with std. dev. σ).
2. **Compute:** $b = As + e \in \mathbb{Z}_q^m$.
3. **Derive state:** $S_0 = H(b)$, where $H : \mathbb{Z}_q^m \to \mathbb{Z}_q^n$ is a hash-based extractor preserving all $\lceil \log_2 q \rceil = 12$ bits per element of b, implemented via SHA-256 with 2-byte encoding per element.

Remark 2 (No Nonce Required)**.** *Because s is freshly sampled per session, the LWE instance (A, b) is computationally indistinguishable from uniform under Decision-LWE. This randomness is sufficient for session uniqueness without an explicit nonce.*

3.4 Keystream Generation

For $t = 0, 1, 2, \ldots$:

$$z_t = F(S_t) \in \mathbb{Z}_q^k \tag{4}$$

$$S_{t+1} = P \cdot S_t + G(z_t) \in \mathbb{Z}_q^n \tag{5}$$

Diffusion matrix P. $P \in \mathbb{Z}_q^{n \times n}$ is a Cauchy matrix defined by $P_{ij} = (i + j + 1)^{-1} \bmod q$. This matrix is verified to satisfy:

- $\det(P) \neq 0$ in $\mathbb{Z}_q$ (invertibility).
- All sampled square submatrices are nonsingular over $\mathbb{Z}_q$.
- Branch number $\mathcal{B}(P) = n + 1$ (maximum possible).[1]

Feedback function G. $G : \mathbb{Z}_q^k \to \mathbb{Z}_q^n$ is defined by:

$$G(z)_i = \prod_{j=0}^{k-1} z_j^{(i+j+2) \bmod 7+2} + (i + 1) \pmod{q}, \tag{6}$$

for $i = 0, \ldots, n - 1$. The exponents $(i + j + 2) \bmod 7 + 2 \in \{2, 3, 4, 5, 6, 7, 8\}$ introduce degree-8 polynomial nonlinearity. Verified properties:

- Perfect differential uniformity (1000/1000 unique differentials, sampling).
- Zero linearity (0/200 linear cases).
- Complete avalanche effect (100% output change under single-element perturbation).

3.5 Encryption and Decryption

The keystream $\{z_t\}$ is serialized to bytes and used in standard XOR-based encryption:

$$\mathrm{Enc}(m) = m \oplus \mathrm{KS}, \qquad \mathrm{Dec}(c) = c \oplus \mathrm{KS}, \tag{7}$$

where KS denotes the byte-serialized keystream.

[1] Verified via SageMath: branch number $= n + 1$ confirmed by exhaustive 1-sparse analysis and 500-sample 2-sparse analysis; all 200 sampled square submatrices of sizes $2, 4, 8, 16$ are nonsingular over $\mathbb{Z}_q$.

4 Security Analysis

4.1 The BF-LWE Problem

Definition 3 (Decision-BF-LWE)**.** *Let n, m, q, χ be parameters and $T \leq 2^{64}$ a query bound. Define two distributions:*

- $\mathcal{D}_{\text{real}}$*: Sample $s \leftarrow \mathbb{Z}_q^n$, $A \leftarrow \mathbb{Z}_q^{m \times n}$, $e \leftarrow \chi^m$. Set $b = As+e$, $S_0 = H(b)$. Generate $z_0, \ldots, z_T$ via the BLiSS keystream equations* (4)–(5)*. Output $(A, b, z_0, \ldots, z_T)$.*
- $\mathcal{D}_{\text{rand}}$*: Sample same (A, b) but output $(A, b, u_0, \ldots, u_T)$ with each $u_t \leftarrow \mathbb{Z}_q^k$ uniform and independent.*

Decision-BF-LWE$_{n,q,\chi,T}$ *is hard if no PPT adversary $\mathcal{A}$ distinguishes $\mathcal{D}_{\text{real}}$ from $\mathcal{D}_{\text{rand}}$ with non-negligible advantage in λ.*

Definition 4 (Search-BF-LWE)**.** *Given $(A, b, z_0, \ldots, z_T)$ sampled from $\mathcal{D}_{\text{real}}$, find the secret vector $s \in \mathbb{Z}_q^n$.*

Remark 3. *The Decision and Search variants are related: any adversary solving Search-BF-LWE trivially solves Decision-BF-LWE. The converse reduction follows from standard LWE self-reducibility arguments [2].*

4.2 Security Reduction

Claim 1. Decision-BF-LWE$_{n,q,\chi,T} \leq_p$ Decision-LWE$_{n,q,\chi}$ *for any $T \leq 2^{64}$. That is, if Decision-LWE is hard and F is perfect nonlinear, then Decision-BF-LWE is hard.*

Proof sketch. We proceed via a hybrid argument over three games.

Game 0 is the real BF-LWE game: the adversary sees $(A, b, z_0, \ldots, z_T)$ where $b = As + e$ is a genuine LWE sample.

Game 1 replaces b with a uniformly random $u \leftarrow \mathbb{Z}_q^m$. By Decision-LWE hardness, no PPT adversary distinguishes Game 0 from Game 1 with non-negligible advantage. Consequently, $S_0 = H(u)$ is computationally indistinguishable from a uniformly random initial state.

Game 2 replaces each z_t with $u_t \leftarrow \mathbb{Z}_q^k$ uniform. With S_0 uniform and F both perfect nonlinear and empirically uniform over $\mathbb{Z}_q^k$ (verified via mean and variance tests at 3σ confidence across all $k = 16$ output components; see Section 5), each output $z_t = F(S_t)$ is statistically uniform over $\mathbb{Z}_q^k$. By the MDS property of P and nonlinearity of G, the state sequence $\{S_t\}$ maintains uniformity across rounds. Hence Game 1 and Game 2 are statistically close, and Game 2 is exactly $\mathcal{D}_{\text{rand}}$ by construction.

Combining: any adversary distinguishing $\mathcal{D}_{\text{real}}$ from $\mathcal{D}_{\text{rand}}$ can be used to break Decision-LWE, contradicting the hardness assumption. □

Remark 4. *A complete formal proof requires: (i) an explicit bound on the statistical distance between Game 1 and Game 2 as a function of T and the LWE parameters, and (ii) a formal proof that the Bent-Induced PRG lemma holds for the MM construction in Equation* (3)*. Both are identified as primary targets for future work.*

4.3 Concrete Security Estimate

Using the LWE estimator of Albrecht et al. [5], we evaluate the concrete hardness of the underlying LWE instance with parameters $(n, q, \sigma) = (460, 3329, 2.0)$ against all standard lattice attacks. Table 1 summarizes the results.

Table 1: Concrete security of BLiSS LWE instance ($n = 460, q = 3329, \sigma = 2.0$) under the LWE estimator [5].

Attack	**Cost (bits)**
uSVP (BKZ reduction)	146.13
BDD (decoding)	142.37
Dual attack	152.85
Dual hybrid	150.00
BKW	167.31
Effective security (minimum)	**142.37**

The effective security of 142.37 bits exceeds the 128-bit post-quantum security target recommended for long-term applications. The modulus $q = 3329$ is a well-studied NTT-friendly prime whose security properties at this parameter level are well-established in the lattice cryptography literature [5].

4.4 Resistance to Known Attacks

Algebraic attacks. The polynomial degree of F is ≥ 3 in all variables (Equation (3)); G achieves degree 8. Low-degree algebraic attacks (e.g., XL algorithm, Gröbner basis methods) require the system degree to fall below the number of unknowns, which is not met for these parameters.

Differential attacks. Perfect nonlinearity of F gives differential uniformity $\delta(F) = q^{-k}$ (the theoretical minimum). Combined with the MDS branch number of P, differential trails over multiple rounds yield no exploitable bias.

Correlation and linear attacks. By the bent property, F admits no nonzero linear approximation; hence correlation attacks find no exploitable statistical bias in the keystream.

Classical key search. Key material is derived from an LWE instance of dimension $n = 460$ over $\mathbb{Z}_{3329}$; exhaustive search requires $\Omega(q^n)$ operations, which is computationally infeasible.

Quantum attacks (Grover's algorithm). Grover's algorithm [1] provides a quadratic speedup for unstructured search. The effective quantum security level is governed by LWE hardness rather than symmetric key size alone. For the chosen parameters, the best known quantum attacks on LWE via quantum lattice sieving do not reduce security below 100 bits; see [5] for concrete estimates.

State recovery. Inverting the state update $S_{t+1} = PS_t + G(z_t)$ requires inverting G, which is protected by its degree-8 nonlinearity. Furthermore, the initial state S_0 is derived from an LWE instance, providing an additional hardness layer against state recovery from observed keystream blocks.

5 Computational Verification

All cryptographic properties were verified using a SageMath prototype. Table 2 summarizes the results.

Output uniformity of F. Since perfect nonlinearity does not automatically imply uniform output distribution, we verify this property independently. For each of the $k = 16$ output components of F, we collect 5,000 samples and test whether the empirical mean and variance are consistent with Uniform($\mathbb{Z}_q$), for which the theoretical values are $\mu = (q-1)/2 = 1664$ and $\sigma^2 = (q^2-1)/12 \approx 923{,}520$. All 16 components pass the 3σ confidence test for both mean and variance, with observed means in the range $[1642, 1690]$ and variances in the range $[906{,}729, 948{,}448]$. This confirms that F is both perfect nonlinear and empirically uniform over $\mathbb{Z}_q^k$, directly supporting the statistical argument in the proof sketch of Claim 1.

Performance. On a standard laptop using unoptimized SageMath: initialization takes $\sim$14.5 ms and keystream generation achieves $\sim$3.5 ms per 32-byte block. These figures are expected to improve substantially with an optimized C or Rust implementation using NTT-based arithmetic over $\mathbb{Z}_q$.

Statistical randomness tests. Keystream quality was evaluated in two stages. First, five internal tests on 10^4–10^5 bits were performed: Frequency (monobit), Runs, Autocorrelation (lags $d \in \{1, 2, 4, 8, 16, 32, 64, 128\}$), Chi-square byte distribution, and Shannon entropy. All tests passed with p-values ≥ 0.01; Shannon entropy reached 7.982 bits/byte (99.77% of the theoretical maximum of 8).

Table 2: Computational verification of BLiSS components (SageMath prototype, $n = 460$, $q = 3329$, $\sigma = 2.0$).

Component	**Property**	**Result**	**Status**
F (Bent function)	Differential uniformity	500/500 unique diffs	**Pass**
F (Bent function)	Max differential collision	1	**Pass**
F (Bent function)	Output coverage	100% of $\mathbb{Z}_{3329}$	**Pass**
F (Bent function)	Avalanche effect	100%	**Pass**
F (Bent function)	Output mean (all k comp.)	≈ 1664 $(= (q-1)/2)$	**Pass**
F (Bent function)	Output variance (all comp.)	$\approx 923{,}520$	**Pass**
F (Bent function)	Uniformity (3σ test)	16/16 components	**Pass**
H (State init)	Full LWE bit preservation	12 bits/element	**Pass**
H (State init)	Collision resistance	None detected	**Pass**
P (Diffusion)	Invertibility	$\det(P) \neq 0$	**Pass**
P (Diffusion)	MDS submatrix property	0/200 singular	**Pass**
P (Diffusion)	Branch number	$n + 1$ (optimal)	**Pass**
G (Feedback)	Differential uniformity	1000/1000 unique	**Pass**
G (Feedback)	Linearity	0/200 linear cases	**Pass**
G (Feedback)	Avalanche effect	100%	**Pass**
Keystream	Block uniqueness (100 rds)	100/100 unique	**Pass**
Keystream	Bit balance	$\pi \approx 0.500$	**Pass**
Keystream	NIST SP 800-22 (15 tests)	All passed, $\geq 8/10$	**Pass**
Keystream	NIST SP 800-22 bitstreams	10×10^6 bits	**Pass**
Keystream	TestU01 SmallCrush	14/15 statistics passed	**Pass**
Keystream	SmallCrush total CPU time	21.1 hours	—
Keystream	SmallCrush WeightDistrib	$p = 3.7 \times 10^{-5}$	**Suspicious**
System	Encrypt/decrypt correctness	Verified	**Pass**
System	Key sensitivity	Distinct keystreams	**Pass**

Second, a full NIST SP 800-22 evaluation was performed on 10×10^6 bits of keystream output (10 independent bitstreams of 10^6 bits each). All 15 statistical tests passed, with proportion scores meeting or exceeding the minimum threshold of 8/10 sequences. P-values were uniformly distributed across all tests, confirming the absence of detectable statistical bias in the BLiSS keystream.

TestU01 SmallCrush results. To complement the NIST evaluation, the BLiSS keystream was subjected to the TestU01 SmallCrush battery [9], which applies 10 families of statistical tests totaling 15 individual statistics. BLiSS passed 14 of the 15 statistics, with only the WeightDistrib test returning a borderline p-value of 3.7×10^{-5} (outside the $[0.001, 0.999]$ interval but well above the failure threshold of $\varepsilon < 10^{-300}$). This slight deviation in

Hamming-weight distribution is characteristic of highly structured lattice-based generators and does not indicate a statistically exploitable bias. The full SmallCrush run consumed 21.1 hours of CPU time on a standard laptop, processing approximately 6.5 billion keystream bits across all tests. Table 2 summarizes the TestU01 results alongside the other verification outcomes.

Table 3: Summary of TestU01 SmallCrush results for BLiSS keystream ($n = 460$, $q = 3329$, $\sigma = 2.0$).

Test	p**-value**	**Status**
BirthdaySpacings	0.75	Pass
Collision	0.28	Pass
Gap	0.03	Pass
SimpPoker	0.41	Pass
CouponCollector	0.13	Pass
MaxOft (Chi^2)	0.13	Pass
MaxOft (AD)	0.09	Pass
WeightDistrib	3.7×10^{-5}	Suspicious
MatrixRank	0.75	Pass
HammingIndep	0.06	Pass
RandomWalk1 (H)	0.34	Pass
RandomWalk1 (M)	0.89	Pass
RandomWalk1 (J)	0.68	Pass
RandomWalk1 (R)	0.54	Pass
RandomWalk1 (C)	0.62	Pass
Total CPU time	21.1 hours	

6 Comparison with Related Work

Table 4: Comparison of BLiSS with existing stream ciphers.

Property	**BLiSS**	**AES-CTR**	**ChaCha20**	**Kreyvium**
Security basis	LWE (formal)	Empirical	Empirical	LWE (partial)
Post-quantum	Yes	Partial	Partial	Yes
Nonce required	No	Yes	Yes	Yes
Nonlinearity	Perfect (bent)	High (S-box)	ARX (heuristic)	Bent filter
Output uniformity	Verified	Assumed	Assumed	Partial
Provable reduction	Yes (sketch)	No	No	Partial
Domain	$\mathbb{Z}_q$	$\mathbb{F}_2$	$\mathbb{Z}_{2^{32}}$	$\mathbb{F}_2$
Concrete security	142.37 bit	≈ 128 bit	≈ 128 bit	≈ 128 bit

Kreyvium [6] is the closest related work, combining a Trivium-like structure with a bent Boolean filter for FHE-friendly evaluation. BLiSS differs in three key aspects: it operates natively over $\mathbb{Z}_q$ rather than $\mathbb{F}_2$, uses LWE for state initialization, and targets a formal security reduction to Decision-LWE rather than FHE efficiency. The SPRING construction [7] also uses LWE for PRF design, but targets a pseudorandom function setting rather than a stream cipher with perfect nonlinearity. BLiSS shares the NTT-friendly modulus $q = 3329$ with well-studied lattice constructions [8], which provides confidence in the concrete hardness analysis of the underlying LWE instance.

7 Conclusion and Future Work

We presented BLiSS, a stream cipher combining Maiorana–McFarland generalized bent functions over $\mathbb{Z}_q$ with LWE-based state initialization. The construction achieves perfect nonlinearity, verified output uniformity, verified MDS diffusion, and a security argument reducing Decision-BF-LWE to Decision-LWE. With parameters $(n, q, \sigma) = (460, 3329, 2.0)$, BLiSS achieves 142.37 bits of concrete security against all known lattice attacks, exceeding the 128-bit post-quantum security threshold. All core components have been verified computationally via SageMath, with results summarized in Table 2.

Future work.

- **Formal security proof.** Complete the Bent-Induced PRG lemma and provide a tight security reduction with explicit statistical distance bounds.
- **Formal uniformity proof.** Establish a theoretical proof that the MM construction over $\mathbb{Z}_q$ yields a uniformly distributed output, complementing the empirical verification reported here.
- **Optimized implementation.** C/Rust implementation with NTT-based arithmetic over $\mathbb{Z}_q$ for practical throughput evaluation.
- **Side-channel resistance.** Evaluation of timing and power side-channel attack resistance.
- **Extended statistical testing.** Following the NIST SP 800-22 (15/15 passed) and TestU01 SmallCrush (14/15 passed) evaluations reported in Section 5, future work will extend the statistical validation to the more intensive TestU01 BigCrush battery for deeper empirical coverage.
- **Formal verification.** Machine-checked proofs of core lemmas.

Acknowledgements

The author acknowledges the use of generative AI tools for assistance in editing and rephrasing certain sections of the manuscript for clarity, as well as for assistance in debugging and optimizing the SageMath and C code used for empirical verification. All intellectual contributions, including the cipher design, the BF-LWE problem, the security proofs, and the analysis of results, are the original work of the author.

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